MAKHANLAL CHATURVEDI NATIONAL UNIVERSITY OF JOURNALISM AND COMMUNICATION, BHOPAL

INFOPARK – THE COLLEGE OF IT AND MANAGEMENT

PROJECT REPORT

MACHINE LEARNING

CAR SPEED PREDICTION

Using Linear Regression

Programme / Class	BCA
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Submitted in partial fulfilment of academic requirements.

Project Title: **Car Speed Prediction Using Linear Regression**

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1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions. It is used in various fields, including healthcare, finance, transportation and automobile technology.

This project, **Car Speed Prediction Using Linear Regression**, demonstrates how a Machine Learning model can estimate the speed of a vehicle using selected vehicle parameters.

The project uses Python and Scikit-learn to train a Linear Regression model. Vehicle-related inputs such as engine RPM, throttle position, brake pressure, engine load, gear and road slope are used to predict vehicle speed in kilometres per hour (km/h).

The project demonstrates the basic Machine Learning workflow, including dataset preparation, model training, prediction and evaluation.

Note: *The project uses synthetic demonstration data. Its results demonstrate the model-building process and do not represent validated predictions from real vehicles.*

2. Problem Statement

Vehicle speed can be related to several parameters, including engine RPM, throttle position, braking, engine load, gear and road slope.

The objective of this project is to develop a Machine Learning model that learns relationships between these input parameters and vehicle speed.

The trained model predicts an estimated vehicle speed from the supplied inputs. Its performance is measured by comparing predicted values with actual target values in the test dataset.

3. Objectives

The main objectives of this project are:

- To understand the fundamentals of Machine Learning and Linear Regression.
- To prepare and explore a vehicle-related dataset.
- To identify suitable input features and the target variable.
- To divide the dataset into training and testing sets.
- To train a Linear Regression model using Scikit-learn.
- To predict vehicle speed from vehicle parameters.
- To evaluate the model using MAE, MSE, RMSE and R^2 score.
- To visualize the relationship between vehicle parameters and speed.

4. Tools and Technologies

Python: Used to write and execute the Machine Learning program.

Pandas: Used to load, inspect and manipulate the dataset.

NumPy: Used for numerical operations.

Matplotlib: Used to create graphs and visualizations.

Scikit-learn: Used for splitting data, training the Linear Regression model and evaluating predictions.

Google Colab / Jupyter Notebook: Used as the development environment for executing the project.

CSV: Used to store the dataset in tabular form.

5. Dataset Description

The project uses a synthetic vehicle dataset containing **20,000 records** and **14 columns**. The dataset is designed for demonstrating the relationship between vehicle parameters and vehicle speed.

The selected model inputs are:

Feature	Description
Engine RPM	Engine rotational speed
Throttle Position (%)	Percentage of throttle input
Brake Pressure (%)	Brake pressure percentage
Engine Load (%)	Engine load percentage
Gear	Selected transmission gear
Road Slope (%)	Road gradient

Target Variable: vehicle_speed_kph — The target variable represents the vehicle speed in kilometres per hour.

Before model training, the dataset is loaded and examined. Its structure, column names, data types and values are checked to ensure that the selected inputs and target are appropriate for the model.

Because the dataset is synthetic, it should not be treated as a substitute for real-world vehicle sensor data.

6. Methodology

The project follows these steps:

Step 1 — Import Libraries: Import Pandas, NumPy, Matplotlib and the required Scikit-learn modules.

Step 2 — Load the Dataset: Read the CSV dataset into a Pandas DataFrame.

Step 3 — Explore the Data: Inspect the rows, columns and relevant vehicle parameters.

Step 4 — Select Features and Target: Select six vehicle parameters as input features and vehicle_speed_kph as the target.

Step 5 — Split the Dataset: Divide the data into training and testing sets using an 80:20 split.

Step 6 — Train the Model: Fit the Linear Regression algorithm using the training data.

Step 7 — Make Predictions: Generate vehicle-speed predictions for the test data.

Step 8 — Evaluate the Model: Calculate MAE, MSE, RMSE and R^2 score.

Step 9 — Visualize Results: Compare actual and predicted speeds and examine prediction errors through graphs.

7. Algorithm: Linear Regression

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values. It learns a mathematical relationship between input variables and a target variable.

For this project, the general multiple Linear Regression equation is:

Predicted Speed = Intercept + (Coefficient■ × Engine RPM) + (Coefficient■ × Throttle Position) + (Coefficient■ × Brake Pressure) + (Coefficient■ × Engine Load) + (Coefficient■ × Gear) + (Coefficient■ × Road Slope)

The model learns the coefficients and intercept from the training dataset. After training, it uses the supplied input values to estimate vehicle speed. The quality of these predictions is evaluated using the test dataset.

8. Results and Evaluation

The model was evaluated on a held-out test set containing **4,000 records**.

The following metrics were obtained from the project:

Evaluation Metric	Result
Mean Absolute Error (MAE)	3.6541 km/h
Mean Squared Error (MSE)	21.1736
Root Mean Squared Error (RMSE)	4.6015 km/h
R^2 Score	0.9647

The R^2 score of **0.9647** indicates that the model explains approximately **96.47%** of the variation in the test target values for this synthetic dataset.

The MAE and RMSE describe the prediction error in speed units. The MSE represents the average squared error.

These results indicate a strong fit to the synthetic test data. They do not establish equivalent accuracy on real vehicles or real-world driving conditions.

9. Example Prediction

Once the model is trained, a user can provide values for the required vehicle parameters:

- Engine RPM
- Throttle position
- Brake pressure
- Engine load
- Gear
- Road slope

The model processes these inputs and returns an estimated vehicle speed in km/h.

Example workflow:

1. Enter the required vehicle parameters.
2. Pass the values to the trained model in the correct feature order.
3. Generate the predicted speed.
4. Display the prediction in km/h.

The exact numerical example prediction should be copied from the executed notebook output so that the report matches the actual implementation.

10. Model Evaluation Notes

Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted speed. A lower value generally indicates smaller errors.

Mean Squared Error (MSE): Measures the average squared difference between actual and predicted speed. Larger errors have a greater effect on this metric.

Root Mean Squared Error (RMSE): The square root of MSE. It expresses error in the same units as the target, which is km/h.

R^2 Score: Measures how well the model explains variation in the target values on the evaluated dataset.

All metrics should be considered together. A strong evaluation result on synthetic data does not guarantee reliable performance under real driving conditions.

11. Advantages

- Demonstrates the basic Machine Learning workflow.
- Uses a clear and understandable regression algorithm.
- Shows how multiple vehicle parameters can be used for prediction.
- Provides numerical metrics for evaluating model performance.
- Includes visual comparisons of actual and predicted values.
- Can serve as a foundation for more advanced vehicle-data projects.

12. Limitations

- The dataset is synthetic rather than collected from real vehicles.

- Real vehicle speed depends on additional factors that may not be included in the model.
- Linear Regression may not capture complex nonlinear relationships.
- Performance on synthetic data may differ significantly from real-world performance.
- The prediction should not be used as a safety-critical vehicle control system without extensive validation.

13. Future Scope

The project can be improved in the following ways:

- Collect real vehicle sensor data for training and testing.
- Compare Linear Regression with other regression algorithms.
- Develop an interactive interface for entering vehicle parameters.
- Add more relevant features where reliable data is available.
- Test the model on different driving conditions.
- Evaluate the model on independent real-world datasets.

14. Conclusion

The **Car Speed Prediction Using Linear Regression** project demonstrates how Machine Learning can estimate a continuous numerical value from multiple input features.

The project covers dataset preparation, feature selection, training and testing, prediction, visualization and model evaluation.

The model achieved an MAE of **3.6541 km/h**, an RMSE of **4.6015 km/h** and an R^2 score of **0.9647** on the synthetic test dataset.

The project provides a practical introduction to regression and demonstrates how a trained model can learn patterns from data. Further work using real vehicle data would be required to assess its usefulness in practical automotive applications.

15. References

1. Scikit-learn documentation — Linear Regression and model evaluation.
2. Pandas documentation — Data handling and analysis.
3. NumPy documentation — Numerical computing.
4. Matplotlib documentation — Data visualization.
5. Python documentation — Programming language reference.
6. Project dataset: Synthetic Car Speed Prediction dataset created for this demonstration.
7. Project notebook: Car Speed Prediction Using Linear Regression.

16. Declaration

I hereby declare that the project report titled **“Car Speed Prediction Using Linear Regression”** has been prepared as part of my academic work for the Machine Learning subject.

The report describes the project methodology, implementation and evaluation. The dataset is synthetic, and the results are presented as obtained from the project’s test evaluation.

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